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Explaining spatial variation and catchment characteristics thresholds in streamflow response

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ABSTRACT

Understanding hydrological processes, considering the interactions among system state, structure, and climate, and the thresholds that regulate catchment behavior, remains a challenge, mainly across scales where multiple processes interact. Here, we move beyond average feature rankings to identify where and when 16 climate and catchment characteristics change their effect on streamflow across 735 Brazilian catchments. Using random forest models together with global and local feature attribution methods, we identify nonlinear thresholds behavior across the flow-duration curve. Model performance supported diagnostic interpretation, with the main streamflow signatures (Q_5 , Q_{mean} , and Q_{95}) showing R^2 and NSE values of 0.63–0.81 and KGE values of 0.70–0.85, while the rarest conditions showed lower but still informative skill. Although these relationships are not causal, provide valuable insights into hydrologic functioning based on the attribute–streamflow relationships identified within the model. Modeled relations indicate that climate exerts the strongest large-scale control: an aridity threshold near 1.23–1.37 marks where drying begins to reduce streamflows, though this effect varies regionally; aridity can enhance streamflow in parts of the Amazon and southern Brazil. Low streamflows were particularly sensitive to precipitation seasonality (SI), which showed a coherent threshold near $SI = 0.40$, as well as to subsurface porosity, slope, elevation, and vegetation cover. Forest cover showed limited relevance for higher streamflows, but low streamflows decline when forest cover exceeds 25%, and a weaker 32–40% threshold appears for Q_{mean} and higher streamflows. Sand content shows a dual behavior with > 56% boosts low streamflows, whereas values above 45% attenuated high streamflows. Regionally, for the rarest high-streamflow (Q_{99}), soil properties can override climate; under shallow bedrock they become the dominant amplifiers. Overall, at large scales, Brazilian streamflow regimes seem organized by a set of physically interpretable thresholds that distinguish climate-dominated regimes from those governed by catchment state and structure, with variation across space and the flow-duration curve.

1. Introduction

Changes in catchment characteristics, including both external forcings (climate, land-use and land cover (LULC), anthropogenic) and catchment state and structure (topography, geology, soil), reorganize runoff-generation and storage processes, yielding nonlinear responses in streamflow regimes. Streamflow combines surface runoff and baseflow, and the flow-duration curve (FDC) summarizes distinct regimes through

percentiles (Burgan & Aksoy, 2022). In practice, the FDC is used with percentile thresholds to delineate flood and low streamflows regimes (Smakhtin, 2001). Therefore, Q_5 is used to represent low-streamflow conditions and with Q_1 capturing the rarer and more volatile tail (e.g., Beck et al., 2015; Johnson et al., 2023; Botterill & McMillan, 2023), while Q_{95} is widely used to denote high streamflows/floods (e.g., Frei et al., 2015; Rahimi et al., 2021; Mitheu et al., 2023). Further, Q_{99} is associated with rare, high-impact events that exceed the 99th percentile

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only under exceptional circumstances (Sharma et al., 2018; Mitheu et al., 2023; Tsiokanos et al., 2024, and references therein), and Q_{mean} summarizes the long-term water balance. Very rare percentiles entail greater statistical uncertainty (Frei et al., 2015; Tsiokanos et al., 2024), yet these tails are where departures from typical hydrologic behavior manifest most clearly. Crucially, the occurrence and magnitude of these percentiles are not linearly related to external forcings; they are co-determined by catchment state & structure (Sharma et al., 2018; Livneh et al., 2024).

Large-sample hydrology (LSH) has standardized datasets and enabled robust, multi-catchment inference on streamflow signatures (e.g., CAMELS-style resources, CABra, Caravan; Addor et al., 2019). Across regions, climate, especially aridity and precipitation, emerges as a dominant control on percentile streamflows, yet findings diverge on the role of state and structure (geology, soils, subsurface storage), particularly for low streamflow and baseflow (Beck et al., 2015; McMillan, 2020; Almagro et al., 2024; Chagas et al., 2024; Correa et al., 2025; Holt & McMillan, 2025). In tropical catchments, spanning humid to semi-arid climates and sharp gradients in soils, relief, and network configuration, signatures seldom respond in the same direction to a given predictor; the hierarchy between climate and catchment properties is signature- and scale-dependent. Given that simple averages can mask nonlinearity and interactions, observations from different regions point to low streamflows being potentially more sensitive to vegetation and storage (Smakhtin, 2001; Filoso et al., 2017; Chen et al., 2023), while high streamflows are often influenced by episodic hydraulic connectivity and storm structure (Sharma et al., 2018; Wasko et al., 2020; Blöschl, 2022; Song et al., 2024); making threshold mapping at the national scale a prerequisite to reconcile seemingly conflicting results and inform management.

Flood occurrence reflects a mix of atmospheric drivers and catchment conditions whose joint effects remain under-explored: many studies stress precipitation variability while under-representing antecedent wetness (Livneh et al., 2024), catchment size and geometry (Sharma et al., 2018; Paiva & Lima, 2024), rainfall organization (Sharma et al., 2018; Tsiokanos et al., 2024), and LULC change (Sharma et al., 2018). For extreme streamflows, evidence conflicts on whether climatic or state and structure controls prevail, and unresolved scale issues persist (Chagas et al., 2024; Pham and Alila, 2024; Nearing, 2020). Machine learning (ML) approaches, such as random forests (RF) (Breiman, 2001), have been increasingly used to analyze these signatures (Almagro et al., 2024; Correa et al., 2025; Holt & McMillan, 2025).

Despite abundant case studies, knowledge about drivers of percentile streamflows remains fragmented and region-specific: catchment attributes that correlate with higher low streamflows in one region may correlate with lower low streamflows elsewhere, limiting generalization (Gnann et al., 2021). Large-sample studies confirm strong climatic controls, but diverge on the role of state & structure, especially for low streamflows and rare high-tails (Beck et al., 2015; McMillan, 2020; Almagro et al., 2024; Chagas et al., 2024; Holt & McMillan, 2025). For Brazil, there is no standardized national-scale LSH assessment that jointly evaluates many attributes and quantifies direction (increase or decrease), magnitude, and spatial-seasonal heterogeneity across FDC, nor whether effects persist or reverse across signatures. Evidence for thresholds in LULC and other attributes is sparse and context-dependent (paired-catchment syntheses vs. multi-regional findings with weak simple correlations), implying additional controls (e.g., Nóbrega et al., 2017; Garcia et al., 2018; Kayitesi et al., 2022).

The field still leans on bivariate correlations (linear/monotonic) and black-box feature rankings (e.g., permutation feature importance in tree-based ensembles), which do not recover magnitude and sign, nonlinear shapes, interactions, or context-dependent breakpoints and can be biased under correlated or mixed-scale predictors (e.g., McMillan, 2020; Almagro et al., 2024; Correa et al., 2025; Holt & McMillan, 2025). Explainable AI (XAI) is growing, but much of it is post hoc and seldom reports global and local effects with uncertainty, precisely where

tail percentiles (Q_1 , Q_{99}) have the highest sampling and model uncertainty and policy salience (Brandão et al., 2025; Camps-Valls et al., 2025; Husic et al., 2025; Slater et al., 2025). For floods, evidence is mixed on climate vs. catchment dominance and hampered by unresolved scale issues; extremes depend on antecedent wetness, event structure, and episodic hydraulic connectivity, not precipitation alone (Sharma et al., 2018; Blöschl, 2022; Pham & Alila, 2024). Together, these knowledge gaps motivate the development a national-scale threshold mapping framework that separates external forcings from state-and-structure controls across hydrological signatures and explicitly quantifies uncertainty.

In this study, we quantify how external forcings (climate, LULC, and anthropogenic influences) and catchment state and structure (topography, geology, and soils) jointly shape the flow-duration curve at Q_1 , Q_5 , Q_{mean} , Q_{95} , Q_{99} in the context of a large sample hydrology. This study seeks to: (i) estimate the direction and magnitude of external forcings versus state-structure controls; (ii) identify, where present, physically interpretable thresholds and potential sign reversals in modeled effects, and maps their spatial expression across hydroclimatic regimes; (iii) characterize spatial and seasonal heterogeneity in these controls; and (iv) contrast the determinants of low streamflows versus upper-tail peaks as a function of hydrological processes and scale.

2. Materials and methods

2.1. Study region and data set

We developed this study across continental Brazil, spanning sharp hydro-climatic gradients set by the South American Monsoon System (SAMS) and modulated by the South American Low-Level Jet (SALLJ) and moisture recycling from the Amazon (Martínez & Domínguez, 2014; Montini et al., 2019). According to the Köppen–Geiger classification for Brazil, climates range from tropical (A, 81%) through semi-arid (B, 5%) to subtropical (C, 14%), subdivided into 12 sub-types that reflect strong spatial heterogeneity (Alvares et al., 2013). Mean precipitation is 1,760 mm.yr⁻¹ at national scale, but regional contrasts are large: the Amazon commonly exceeds 2,000 mm.yr⁻¹ with high runoff, whereas parts of the Northeast are water-limited ($P < PET$) with 600–1,200 mm.yr⁻¹ of rainfall and much lower runoff. These gradients, together with strong variations in soils, lithology, relief and river-floodplain morphology, create distinct controls across low, mean, and high-streamflow signatures (Fig. 1a–e), against the country's biome mosaic (Fig. 1f).

We use the Catchments Attributes for Brazil (CABra) dataset (see Almagro et al., 2021 for full documentation), covering 735 gauged catchments with harmonized meteorological forcings, daily streamflow, and an extensive set of static attributes. Streamflow series are compiled from Brazil's national hydrometric network, while catchment descriptors integrate multiple vetted sources—most prominently Hydro-ATLAS for hydrography and topography (Linke et al., 2019), plus widely used global and regional layers for climate, soils, geology, lithology, topography, and LULC. Together, these provide the standardized feature space required for LSH inference on percentile streamflows under Brazil's diverse hydro-climatic regimes.

2.2. Random forest inputs and targets

RF is a supervised learning approach that employs an ensemble of decorrelated decision trees constructed using bootstrap sampling, reducing the likelihood of overfitting (Breiman, 2001). The advantages being its ability to capture non-linear relationships between inputs and outputs by leveraging its data-driven nature to uncover hidden patterns without being constrained by predefined assumptions about input variables, while also providing interpretability by assessing the importance of each predictor based on its contribution to reducing impurity at decision nodes (Breiman, 2001; Addor et al., 2018; Tyrallis et al., 2019). For a detailed explanation of the model's complexity and mathematical

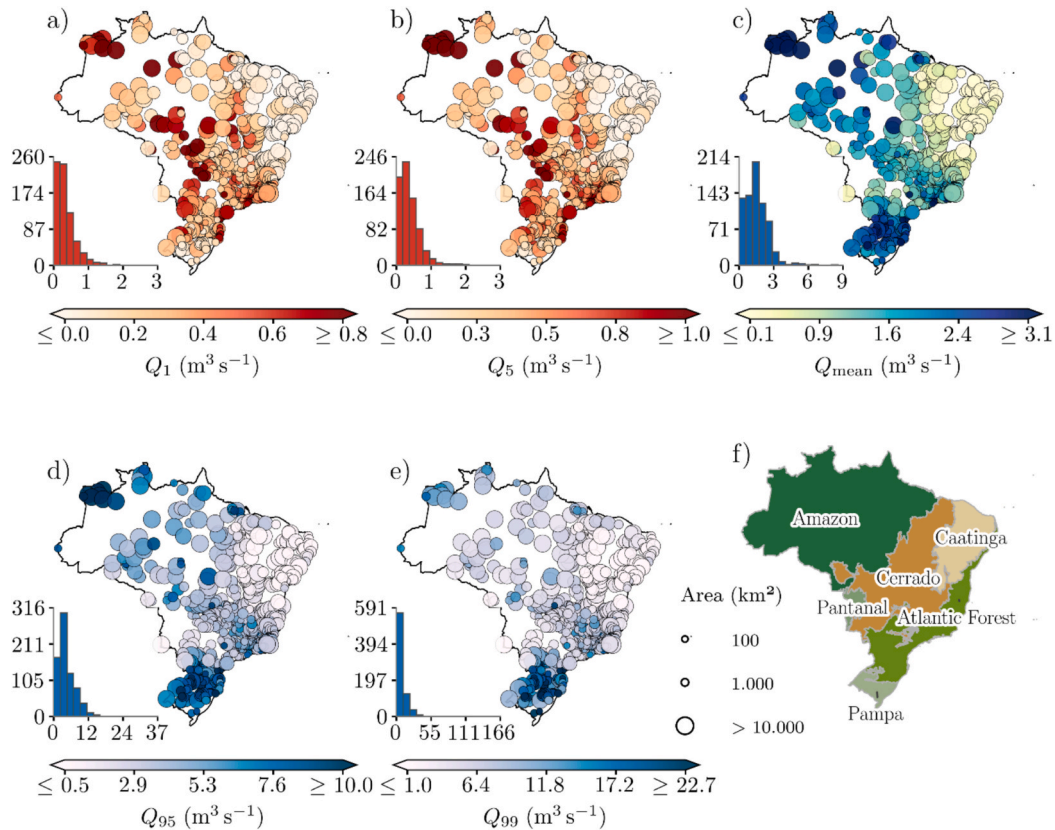


Fig. 1. Spatial distribution of streamflow statistics across Brazil. Panels (a–e) show gauging sites; circle area scales with drainage area, and color encodes ($\text{m}^3 \cdot \text{s}^{-1}$): a) Q_1 , b) Q_5 , c) Q_{mean} , d) Q_{95} , e) Q_{99} . Each panel includes an inset histogram summarizing the distribution of the corresponding statistic. Thin lines indicate internal biome boundaries and panel f) maps Brazilian biomes.

foundations, refer to Breiman (2001).

The RF analyses has become a consolidated tool in multiple geoscience fields, especially in LSH. Here, we used a RF to predict streamflow percentiles: 1st (Q_1), 5th (Q_5), mean (Q_{mean}), 95th (Q_{95}), and 99th (Q_{99}), using climatic and catchment attributes of the catchments. There is no consensus in the literature on universal thresholds for low or high streamflows, and estimates at the distribution tails typically carry greater uncertainty (Frei et al., 2015; Tsiokanos et al., 2024). Still, these tails capture the anomalous hydrologic behavior often linked to climate-change forcing; therefore, we explicitly analyze Q_1 and Q_{99} alongside Q_5 , Q_{mean} , and Q_{95} . We used 16 attributes (Table 1) selected in line with McMillan (2020) and Almagro et al. (2024). A detailed description of the primary catchment characteristics is provided in Table S1, while their spatial distributions and others statistics can be found in Almagro et al. (2021).

2.3. Hyperparameters exploration and sensitivity

Hyperparameters optimization is often critical for the performance of machine learning models and for the stability of inferences derived from flexible models (Sweet et al., 2023; Hosseini et al., 2024). As an alternative to exhaustive grid search, the literature shows that random search is generally more efficient for identifying strong solutions at a much lower computational cost, especially when only a few hyperparameters are truly dominant (Bergstra and Bengio, 2012). Therefore, to assess the robustness of the inferred relationships (e.g., variable importance and thresholds), we adopted a random search strategy combined with repeated refits across multiple random seeds, consistent with recent studies applying random-search optimization in hydrology (e.g., Virro et al., 2022; Hosseini et al., 2024; Hosseini et al., 2025).

Many strategies are available for assessing model generalization,

Table 1

Catchment characteristics external forcings and state & structure.

Class	Attribute	Long name	Unit	Source
Anthropogenic	Disturbance	Hydrological disturbance index	–	Almagro et al. (2021)
Climate	Aridity	Aridity index	–	Almagro et al. (2021)
Geology	Precip. seasonality	Seasonality of the precipitation	–	Almagro et al. (2021)
	HAND	Height above the nearest drainage	–	Nobre et al. (2011)
	Porosity	Subsurface soil porosity	–	Gleeson et al. (2014)
	Permeability	Subsurface soil permeability	–	Gleeson et al. (2014)
Land use and cover	Forest	Forest fraction of cover	%	Buchhorn et al. (2019)
	Bare soil	Bare soil fraction of cover	%	Buchhorn et al. (2019)
	Grass	Grass fraction of cover	%	Buchhorn et al. (2019)
	Summer NDVI	NDVI in DJF (rainy season)	–	Buchhorn et al. (2019)
Soil	Soil sand	The sand fraction in the soil	%	Hengl et al. (2017)
	Soil carbon	The carbon fraction in the soil	%	Hengl et al. (2017)
	Soil depth	Depth of the soil to the bedrock	m	Hengl et al. (2017)
	Topography	Area of the catchment	km^2	Almagro et al. (2021)
Topography	Elevation	Mean elevation of the catchment	m	Almagro et al. (2021)
	Slope	Mean slope of the catchment	%	Almagro et al. (2021)

including resampling and data splitting methods. In this study, we followed the approach described by James et al. (2023) and adopted in hydrological studies (e.g., Virro et al., 2022; Pandit et al., 2025; Husic et al., 2025), in which the primary generalization metric was the out-of-bag (OOB) estimate a type of resampling. This choice is justified because RF is a bootstrap-based bagging model: each tree is trained on a bootstrap sample, leaving on average about one-third of the observations out of the training set for that tree. Accordingly, we did not use a fixed external holdout split or external cross-validation. Instead, models were fitted using all available catchments, and the OOB prediction for each catchment was obtained by aggregating only the trees that did not use that catchment during training, thus providing an internal estimate of test error. With a sufficiently large number of trees, the OOB error is asymptotically equivalent to the leave-one-out cross-validation error. This approach is particularly convenient when the objective is to characterize model relationships and stability while avoiding the additional computational cost of extensive cross-validation. In addition, because our primary goal was to explain patterns and controls rather than solely maximize predictive accuracy, our framework follows the distinction between modeling for explanation versus prediction proposed by Shmueli (2010), using OOB performance as the main evaluation criterion together with additional sensitivity analyses.

We implemented a random search over a discrete hyperparameters space including the main RF hyperparameters, defined in accordance with the literature (Biau and Scornet, 2016; Probst et al., 2019; James et al., 2023; see Table S2 for the hyperparameters space used in the random search). From all possible combinations, we sampled 1,000 unique hyperparameters configurations and trained models using 10 random seeds. To construct the ensemble, configurations were ranked according to the median OOB performance across seeds, using OOB-KGE as the primary ranking metric. Ties were resolved by selecting the configuration with lower variability across seeds, measured by the interquartile range (IQR) (see Table S3). The top 10 configurations for each target variable were retained for interpretability analyses (see Table S4).

Next, each of the top 10 configurations was retrained using 10 random seeds, and OOB metrics were recalculated, resulting in a performance distribution for each target. In this way, the analysis explicitly quantifies sensitivity to model stochasticity (see Table S5). Because RF does not explicitly account for spatial structure, spatial dependence in the data may affect predictive performance in spatial applications (Roberts et al., 2017; Nowosad & Meyer, 2025). We therefore additionally diagnosed residual spatial autocorrelation using the global Moran's I statistic (Moran, 1950), a standard measure of spatial autocorrelation, calculated on OOB residuals using a k-nearest neighbors structure ($k = 8$ and 499 permutations) (Kim et al., 2023; Nowosad & Meyer, 2025). Moran's I values and permutation-based p-values were calculated for each target and repeated across 10 RF refits with different random seeds; summary statistics are presented in Table S6.

For the top 10 retained configurations for each target, the descriptive statistics therefore quantify the variability induced by RF stochasticity, including both bootstrap sampling of observations (bagging) and feature subsampling at split nodes. Model performance was summarized using the coefficient of determination (R^2), mean squared error (MSE), the Nash-Sutcliffe efficiency coefficient (NSE) (Nash and Sutcliffe, 1970), and the Kling-Gupta efficiency (KGE) (Gupta et al., 2009).

2.4. Explaining spatial variation and feature thresholds

RF are widely used to relate catchment attributes to key hydrologic targets, but standard variable-importance summaries do not indicate direction (whether increasing a feature tends to raise or lower the prediction) nor reveal where effects intensify. To address this, we paired fold-aggregated permutation importance with out-of-fold SHAP (SHapley Additive exPlanations), an explainable-AI framework grounded in cooperative game theory (Lundberg & Lee, 2017). SHAP attributes a

model prediction to individual features by the average marginal contribution a feature makes across all possible coalitions of features. Formally, for model f and input x , the Shapley value for feature i is

$$\psi_i(v) = \sum_{S \subseteq D \setminus \{i\}} \underbrace{\frac{|S|!(|D| - |S| - 1)!}{|D|!}}_{S' \text{ weight}} \left[\overbrace{v(S \cup \{i\}) - v(S)}^{i' \text{ marginal contribution}} \right] \quad (4)$$

where, ω_i is the SHAP value for feature i . The summation runs over all subsets S of features, excluding feature i . $|S|$ is the number of features in S , and $|D|$ is the total number of features. $v(S \cup \{i\})$ is the model's predicted value with the features in S plus feature i , and $v(S)$ is the predicted value of the model with just the features in S .

In practice, we used *TreeSHAP* with the interventional approach (Lundberg et al., 2020), which computes Shapley values (ψ) to explain the RF model by assuming feature independence when estimating feature contributions. Equation (4) defines ψ_i , and thus the marginal contribution of feature i to a given a prediction. A positive ψ_i means feature i pushes the predicted percentile up relative to the model's expected value; negative ψ_i pushes it down (i.e., highlighting where external forcings and state & structure potentially act to increase or decrease Q).

Because the interpretation of threshold-like patterns in SHAP dependence curves may itself be influenced by hyperparameters choice and random-seed variation, SHAP analyses were conducted for the top-10 RF hyperparameters configurations retained for each target and repeated across multiple random-seed refits, thereby propagating RF stochasticity into the interpretability results. To summarize potential threshold behavior in SHAP dependence relationships, we estimated a breakpoint for each predictor from its smoothed SHAP dependence curve. Feature values were grouped into 25 quantile-based bins, and the median SHAP value was calculated within each bin to construct a smoothed dependence profile. A candidate threshold was then defined as the first zero crossing of this curve; when no zero crossing was detected, the point of maximum absolute change between adjacent bins was adopted as a conservative alternative. Uncertainty in breakpoint estimation was quantified through nonparametric bootstrap resampling of catchments: for each bootstrap replicate, the breakpoint was re-estimated, yielding a sampling distribution from which we report the median threshold (Fig. S1) and its 95% bootstrap confidence interval (see Fig. S2), thereby indicating where inferred breakpoints are consistently identified versus weakly constrained.

From the ψ_i results we may identify threshold-like behavior beyond which a predictor exerts a markedly stronger (or opposite) influence, indicating potential nonlinear regimes (Brandão et al., 2025; Pandit et al., 2025; Husic et al., 2025). Identifying these thresholds is important in tropical catchments, where small changes in key attributes can shift regimes, supporting the use of interventional ψ to explore their potential effects. For example, given a Cerrado biome catchment with aridity index ($\phi = \overline{E_p}/\overline{P}$) = 1.2 and forest cover = 28%, SHAP lets us ask: i. how would the predicted streamflow percentiles Q change if forest cover increased from 28% to 45%? ii. how would Q change if precipitation seasonality (SI) decreased from a high to an intermediate value? These what-if scenarios are illustrative of model sensitivities and aid interpretation of spatial patterns; they are not causal claims.

3. Results and discussions

3.1. Streamflows estimation and dominant hydrological controls

This study evaluates the climatic and catchment characteristics that influence mean streamflow and the 1st, 5th, 95th, and 99th streamflow percentiles across Brazilian catchments. The main text focus on 5th, mean, and 95th percentiles, highlighting their differences from the 1st and 99th percentiles (details for Q_1 and Q_{99} in the Supplementary

Material). Unless stated otherwise, figures in the main text present results from the best-performing RF configuration for each target, whereas the effects of configuration choice and random-seed variation are summarized separately. For these principal quantiles, across the retained top-10 RF configurations and 10 seed refits per configuration, OOB performance mean was highest for Q_{mean} , followed by Q_{95} and Q_5 , with R^2 ranging from 0.63 to 0.81, MSE from 0.04 to 2.84, NSE from 0.63 to 0.81, and KGE from 0.70 to 0.85 (Table S5). Moreover, the modest train-to-OOB performance gaps across the retained models (Table S3) provide no evidence of substantial overfitting, suggesting that the RF models retained reasonable generalization beyond the bootstrap training samples. These values fall within ranges commonly reported for RF regionalization with static catchment attributes (Addor et al., 2018; Almagro et al., 2024; Correa et al., 2025). In contrast, performance declines for the rare events, for Q_{99} ($R^2 = 0.44$; MSE = 42.88; NSE = 0.44; KGE = 0.54) and Q_1 ($R^2 = 0.63$; MSE = 0.034; NSE = 0.57; KGE = 0.63), consistent with increase in prediction errors toward the tails of the FDC (McMillan et al., 2012; Fouad et al., 2018; Addor et al., 2018). At the same time, given the greater hydrological complexity of these extreme conditions, the observed performance remains acceptable for regional inference. Low-streamflow conditions are seasonally recurrent in many Brazilian catchments, often intensifying toward the end of the dry season (Chagas et al., 2024), which helps explain why they are more predictable than higher streamflows.

Model performance was only weakly sensitive to the choice of random seed for the main targets, as repeated refits showed very limited dispersion in OOB metrics (Table S5). This indicates that the selected RF configurations produced stable results despite the stochasticity inherent to bootstrap resampling and feature subsampling. Residual spatial autocorrelation was generally weak, with Moran's I values close to zero overall, suggesting that spatial autocorrelation likely did not substantially inflate model performance (Table S6). For Q_1 and Q_5 , permutation-based p-values were not significant, indicating little evidence of residual spatial structure, whereas Q_{mean} , Q_{95} , and Q_{99} showed significant but low-magnitude residual spatial autocorrelation. Together, our values are adequate for capturing the explained proportion of variance (R^2), error magnitude (MSE), predictive efficiency relative to observed variability (NSE), and the joint representation of correlation, variability, and bias (KGE), consequently are adequate to support ψ -based interpretation of attribute-response relationships.

For Q_1 and Q_5 , climatic descriptors remain the primary controls: the ϕ ranks first and SI follows (Fig. 2a, S1a and S2a-b), indicating that long-term water availability and the seasonal rhythm of rainfall set the overall low-streamflow level in catchments. For both quantiles, which capture the tail end of the dry-season recession, the results show that catchment properties matter: porosity, slope, and mean elevation are associated with faster depletion of near-surface stores, consistent with classic low-streamflow theory that emphasizes storage-release and recession controls at low streamflows. In an ablation test excluding ϕ , out-of-sample skill did not change significantly, while feature rankings shifted: SI became the leading climatic descriptor for Q_1 and grass cover for Q_5 , and storage and structure of the catchments proxies (soil carbon fraction, forest cover, porosity, slope, and elevation) together rose in rank. This result agrees with Beck et al. (2015) on the relevance of topography and, together with Brazilian evidence from Chagas et al. (2024) showing that geology and soils modulate low-streamflow variability alongside climate, indicates, contrary to Beck et al. (2015), that geology is not among the least informative. Low streamflows are more sensitive to environmental change than high streamflows (Chen et al., 2023). Our results reflect this: in contrast to high streamflows, vegetation controls are stronger, with forest cover and grass cover emerging as important; in such cases, groundwater-river interactions allow vegetation water use to influence baseflow. This pattern is widely reported in empirical studies, which show that increasing vegetation cover commonly reduces water yield and can reduce streamflows, whereas deforestation increases river streamflows (Filoso et al., 2017; Nóbrega

et al., 2017; Levy et al., 2018).

For Q_{mean} , which synthesizes the long-term water balance, the most relevant features after ϕ are soil carbon fraction, area, and bare soil (Fig. 2b, S2b). Higher soil carbon fraction increases plant-available water and infiltration, enhancing storage and recharge (Hudson, 1994; Iltstedt et al., 2016); larger catchment area scales absolute discharge (Pham & Alila, 2024); greater bare-soil fractions promote runoff (Schwambach et al., 2024), and, together with deforestation, which can raise groundwater levels (Houspanossian et al., 2023), can intensify river-aquifer exchange. Unlike low-streamflow signatures, SI does not rank among the top catchment characteristics of Q_{mean} because large spatial contrasts in P and PET dominate the mean streamflow, effectively overshadowing seasonal-distribution effects.

For Q_{95} , the most influential attributes are soil carbon fraction, permeability, and area (Fig. 2c). For Q_{99} , the ranking shifts toward near-surface storage and infiltration capacity: sand fraction leads, followed by ϕ and soil depth, with soil carbon fraction also among the top predictors. In both cases, catchment properties act jointly with climate (ϕ), indicating that high streamflows are modulated by antecedent wetness (soil carbon fraction), transmissivity (permeability), and contributing area (area); the rarest extremes additionally depend on infiltration limits and shallow storage (sand fraction and soil depth). This perspective aligns with Sharma et al. (2018), Ran et al., 2022 and Livneh et al. (2024), who emphasize that flood modulation is not determined solely by increases in precipitation, but also by state and structure of the catchment.

In sum, climate sets the backdrop, but the hydrological outcome is co-determined by catchment state and structure. The long-term water balance (P vs. PET) anchors all signatures; Q_1 – Q_5 emerge when rainfall seasonality is filtered by storage/transmissivity and by the drainage efficiency of the terrain, whereas Q_{99} depends on infiltration limits and shallow soil storage, modulated by antecedent moisture (including influences of soil carbon fraction). Recent experience confirms this view: the historical floods in southern Brazil (2023/2024) show a detectable contribution from warming (Clarke et al., 2024), yet the extreme discharges were modulated by regional physical characteristics (Clarke et al., 2024; Ferreira, 2024), reinforcing the dependence on the interaction between climate forcing and catchment physical properties. At the other end, Espinoza et al. (2024) show that the 2023 Amazon drought broke records for temperature and river levels. A plausible interpretation, consistent with literature (e.g., Apurv et al., 2017; Condon et al., 2020) on heat-moisture coupling and hydrological-drought propagation, is that extreme temperatures intensified vegetation evapotranspiration and accelerated drainage of shallow storage, contributing to the observed minima; although we cannot claim direct causality here, and our results indicate a vegetation signal acting on low streamflows. This reading is consistent with how Q_1 – Q_5 tend to respond to climate-vegetation-storage coupling.

The Amazon also feeds precipitation to the Southern Cone via the SALLJ and moisture recycling, so vegetation changes reverberate in peaks and low-streamflow seasons thousands of kilometers downwind; studies show that deforestation reduce southward moisture transport, weakening the wet season and potentially increasing signature intermittency (Martinez & Dominguez, 2014; Montini et al., 2019; Bottino et al., 2024). This implies that vegetation is not only a local sink that depresses baseflow via ET; it is also an atmospheric source that sustains downstream precipitation. In signature terms, the biosphere can raise Q_5 and Q_{mean} regionally (via recycling) even while reducing Q_1 locally (via ET). This apparent paradox resolves only if we treat climate, catchment, and biosphere as a coupled system; in practice, vegetation trends to the north can alter the regional climatic baseline of the Q's, not only how a catchment transforms rainfall into discharge.

3.2. Spatial patterns of dominant controls on streamflow signatures

Despite consistent continent-scale leaders, the dominant driver is catchment-specific; Fig. 3 (and S2) shows, for each catchment, the

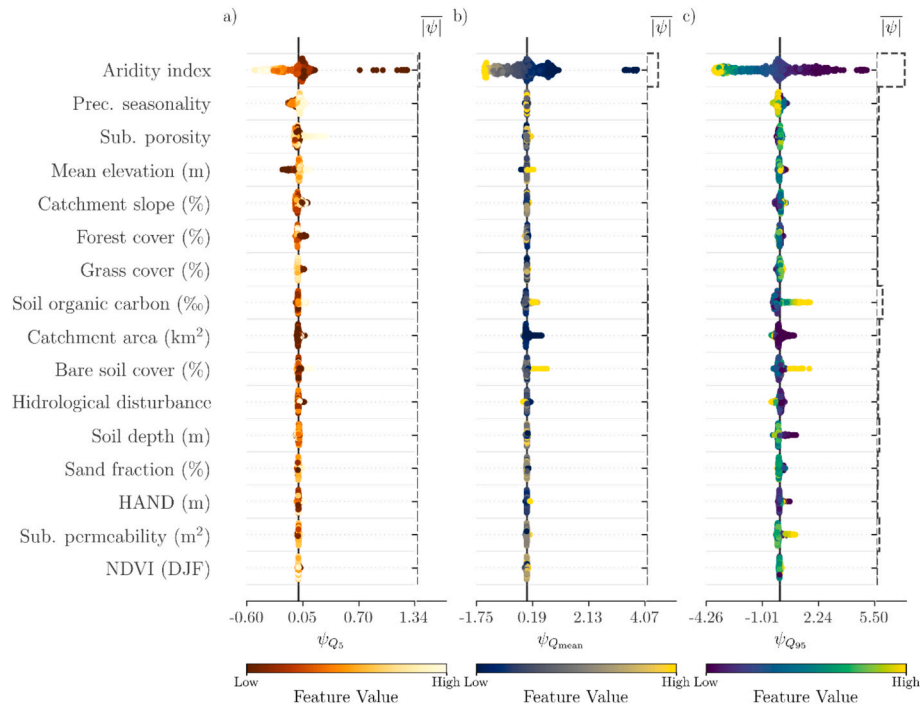


Fig. 2. SHAP summary plots showing how catchment attributes influence Random Forest predictions of Q_5 , Q_{mean} , and Q_{95} across 735 Brazilian catchments. Each point represents one catchment. The horizontal position gives the SHAP value (ψ_i), that is, the contribution of a given attribute to the prediction relative to the model baseline: positive values ($\psi > 0$) increase the predicted streamflow percentile, whereas negative values ($\psi < 0$) decrease it. Points are colored by the relative value of the predictor, from low to high, as indicated by the color bar below each panel. The staircase plot on the right axis indicates the mean absolute SHAP value $|\bar{\psi}|$ of all observations for a predictor.

attribute group with the most $\pm \psi$ (min/max ψ), i.e., the group that most reduces or increases each streamflow signature for each catchment. For low streamflows (Fig. 3a and S3a), climate suppresses Q_5 and Q_1 over the Caatinga biome and parts of the southern Brazil (44% of catchments for Q_5 ; 54% for Q_1), while catchment properties dominate much of central Brazil. This is physically consistent: in water-limited settings where precipitation and evaporative demand are in phase, ϕ and strong seasonality lengthen the dry window and allow storage to drain, i.e., climate is the binding constraint that suppresses minima (Apuv et al., 2017; Soares et al., 2021). In the southern Brazil, the predominance of basaltic bedrock (Serra Geral fractured aquifer) (Vargas et al., 2022), clay-rich soils derived from basalt, and locally shallow regolith (order of meters on slopes) (Chagas et al., 2024) plausibly limits effective storage and low-head transmissivity; nonetheless, the climatic block emerges as the main low-streamflow suppressor even under humid subtropical climatology with relatively low seasonality (Alvares et al., 2013), because spatial climate gradients explain more of the minima than geologic–pedologic contrasts at our analysis scale. By contrast, across the Cerrado and central plateau and the central-Serra Geral belt (sedimentary bedrock; depth to bedrock ≥ 5 m; sand fraction content $\geq 60\%$), subsurface storage and transmissivity (porosity, soil depth, permeability) control how long baseflow can be sustained under an intermediate climate, so that nearly half of the catchments in this belt exhibit physical characteristics that potentially act as hydrological stressors on low streamflows. Conversely, in the Amazon region, flat topography and shallow, unconsolidated substrates promote strong vegetation–river interactions, bolstering storage–discharge connectivity and mitigating low-streamflow suppression. Again, empirical observations support that vegetation type, in concert with geology–soil characteristics, influences streamflow during low-precipitation periods (Bosch & Hewlett, 1982; Istedt et al., 2016; Segura et al., 2019).

Conversely, for increases in low streamflows (Fig. 3d and S3c), the climate block dominates along the Amazon-to-South and Southeast corridor, whereas catchment attributes drive most positive responses

elsewhere. In the Cerrado, increases are chiefly linked to geology and subsurface structure (HAND, porosity, and permeability) which together enhance recharge and transmissivity storage, sustaining baseflow and lifting Q_1 – Q_5 .

For Q_{mean} (Fig. 3b), we identify reductions controlled by catchment characteristics in the South, Southeast, and North, whereas climatic variables dominate Q_{mean} in the Caatinga, parts of the Cerrado, and the northeastern Atlantic Forest. This pattern is consistent with the Budyko framework and with evidence for Brazil: in water-limited domains, the P–PET balance governs water partitioning and thus Q_{mean} , making the climate block the dominant explainer (Budyko-like; P vs. PET). At the humid end-member, dense forests and the hydrodynamics of the Amazon floodplain (storage and backwater effects) lower mean yield by elevating ET and attenuating and delaying discharge propagation. These mechanisms help explain why physical attributes tends to reduce Q_{mean} in the North and in wetter portions of the South and Southeast.

By contrast, increases in Q_{mean} align with the Amazon-to-South/Southeast corridor, where moisture convergence by the SALLJ strengthens precipitation and therefore the climatic mean of streamflow; outside this corridor, catchment attributes account for positive increments. In parts of the Cerrado, geology and subsurface structure (porosity, permeability, and effective depth) favors recharge and transmissive storage, sustaining baseflow and raising Q_{mean} under an intermediate climate. Additionally, studies show that vegetation loss can raise groundwater levels and increase surface runoff, thereby boosting baseflow and the surface-runoff contribution to mean streamflows (Houspanossian et al., 2023), which contextualizes Q_{mean} increases associated with land-cover/use changes. In the Caatinga, LULC tends to enhance streamflow, and soil–plant–atmosphere fluxes are controlled by the upper soil layer (Pinheiro et al., 2017), which modulates soil moisture and evapotranspiration and, in turn, the soil’s storage capacity, recharge, and runoff production—consistent with regional studies showing that LULC changes (Scanlon et al., 2006; Montenegro & Ragab 2010; Lins et al., 2024) and broader anthropogenic land and water use

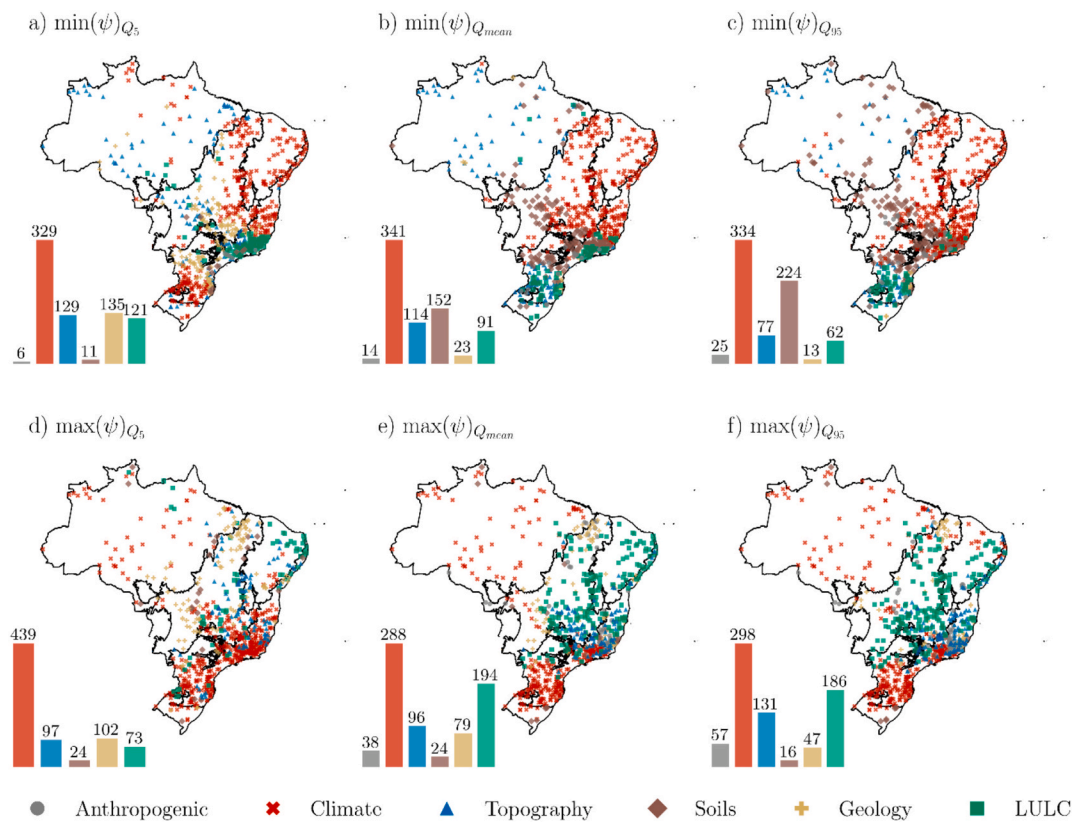


Fig. 3. Spatial distribution of the dominant groups of predictors controlling streamflow signatures across Brazilian catchments. For each catchment, panels (a–c) show the attribute group containing the predictor with the most negative SHAP value ($\min \psi$), that is, the predictor contributing most strongly to decrease the predicted Q_5 , Q_{mean} , and Q_{95} , respectively. Panels (d–f) show the attribute group containing the predictor with the most positive SHAP value ($\max \psi$), that is, the predictor contributing most strongly to increase the predicted Q_5 , Q_{mean} , and Q_{95} . Each environmental group is identified by a unique color and marker shape: Anthropogenic, Climate, Topography, Soils, Geology, and Land Use and Land Cover (LULC). Insets in each panel summarize the number of catchments assigned to each group. Thus, the maps show where different groups of controls dominate the strongest negative or positive contributions to each streamflow signature.

(Ferraz et al., 2025; Zhao et al., 2025) dominate streamflow alterations rather than climate alone.

For Q_{95} (Fig. 3c), climate tends to attenuate Q_{95} over water-limited regions (northern Cerrado and Caatinga biomes). Pronounced seasonality and persistent soil-moisture deficits mean that even intense storms produce muted peaks unless multi-event wetting closes the moisture gap (Tramblay et al., 2010; Sharma et al., 2018; Livneh et al., 2024). Q_{95} and Q_{99} show the same pattern; Livneh et al. (2024) report that Q_{99} is highly sensitive to intermittency and dry spells longer than 20 days, so floods there typically materialize only under the most intense storms.

Across parts of the Serra Geral and Serra do Mar belts that encroach on the Cerrado, soils emerge as the principal modulators of high streamflows (more strongly for Q_{99} ; Fig. S3b). Sandy mantles over locally shallow bedrock juxtaposed with deeper profiles elsewhere, together with elevated relief (1000 m) and, in Atlantic Forest settings, carbon-rich structured soils, may promote infiltration, transmission, and longer hillslope residence times, thereby to damping peak discharge. In the Cerrado band, high-infiltration soils and porous subsurface structure favor subsurface pathways and transient storage; the hydrograph becomes more spread out in time, with both baseflow and quickflow components appearing less sharp.

In the Amazon, topography and soils likewise appear to attenuate maxima. Extremely low gradients, pervasive backwater, and floodplain storage can make flood waves more diffusive, delaying and reducing peaks along the river course and producing broad, long-lived high stages rather than sharp crests (Paiva et al., 2013a, 2013; Paiva & Lima, 2024). Extensive floodplain storage combined with organic-rich, often sand-bearing soils may enhance infiltration and transmissive storage, further buffering extremes. Where the climate block increases Q_{95}

magnitudes, the spatial pattern broadly follows the Amazon-to-South and Southeast corridor, but it involves a smaller fraction of catchments than for Q_{mean} or low streamflows—consistent with the idea that this climatic corridor helps set long-term FDC availability yet exerts a more limited control on peaks than state and structure. For Q_{99} (Fig. S3d), this corridor weakens: climate influence shifts partly toward the central Cerrado, and soil properties there become even more influential for the most extreme streamflows.

3.3. Spatial variability of attributes and thresholds in streamflow response

Unlike previous studies, which have mainly focused on the average importance of each attribute (e.g., Chagas et al., 2024; Almagro et al., 2024; De la Fuente et al., 2025; Holt & McMillan, 2025), our analysis identifies inflection points in the marginal effects of catchment characteristics on streamflow quantiles using ψ dependence curves and spatially distributed ψ values. In a continental, hydrologically diverse country like Brazil, ψ sign maps and their thresholds (Figs. 4–6 for Q_5 – Q_{95} ; Figs. S4 and S13 for all; Tables S7–S9 for threshold uncertainty) indicate that the same predictor may shift from amplifier to attenuator depending on climatic regime and catchment context. These patterns emerge only when the analysis is stratified by key covariates. The curves represent the nationwide interventional mean effect, whereas ψ quantifies the modeled marginal effect contribution under the interventional assumption; it does not imply direct causality. Therefore, we interpret thresholds in the light of physical mechanisms and spatial context, where the literature provides mechanistic plausibility. Bare soil and the anthropogenic disturbance (HDI) do not show stable breakpoints, and their effects remain centered around zero, likely because their

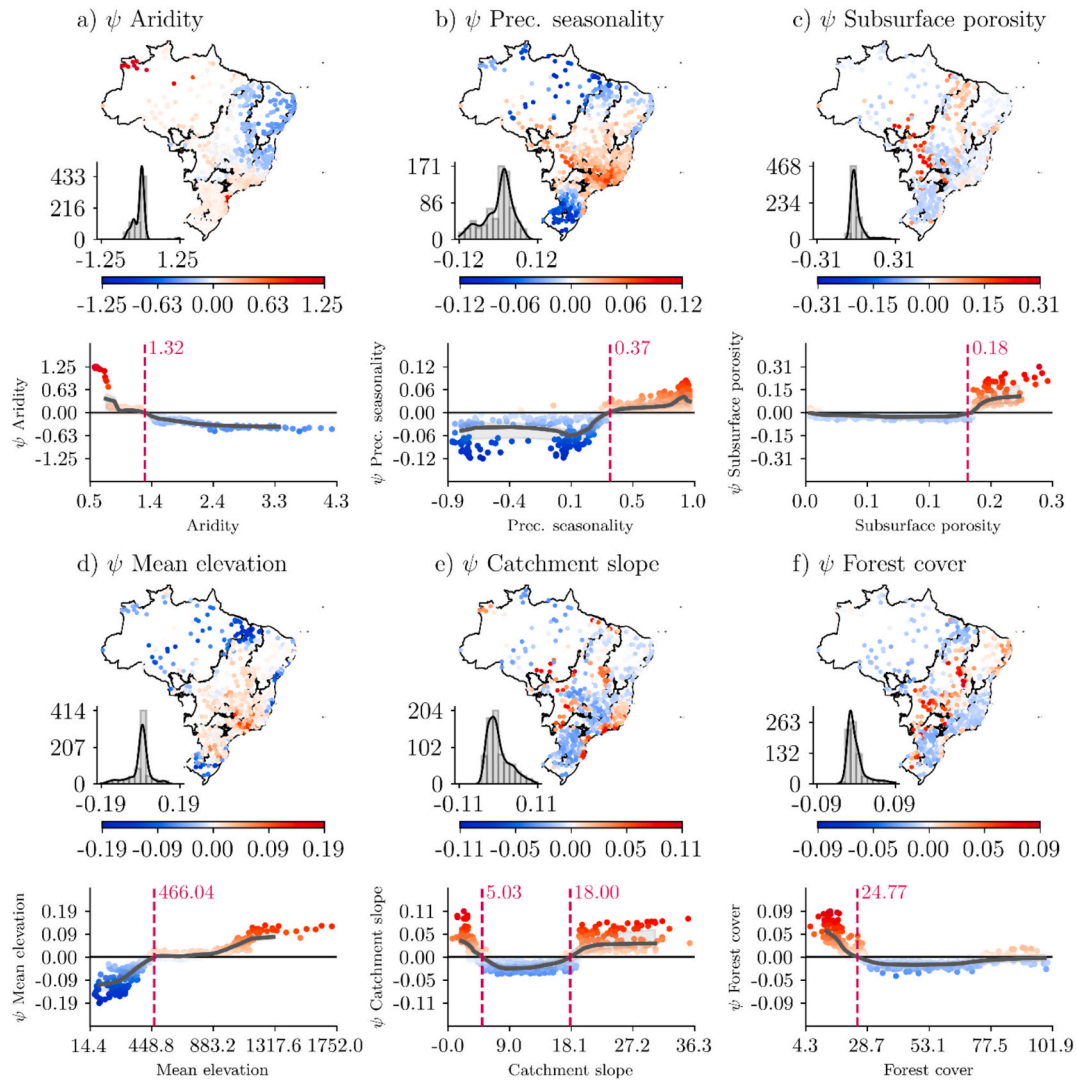


Fig. 4. Spatial SHAP patterns and potential thresholds for the six leading catchment characteristics of Q_5 across Brazilian catchments (remaining predictors in Figs. S5 and S10). For each catchment characteristics, the upper panel shows the spatial distribution of catchment-scale SHAP values (ψ_i), where warm colors indicate positive contributions to predicted Q_5 and cool colors indicate negative contributions; darker colors denote stronger effects. Insets summarize the ψ_i distributions, with $\psi_i = 0$ separating negative from positive effects. The lower panel shows the SHAP dependence plot, in which each point is one catchment, the black curve is the smoothed median response, and the gray band is the interquartile range. Horizontal zero lines mark the switch between negative and positive effects, and vertical dashed magenta lines mark potential breakpoints where the median SHAP response changes sign or slope. Together, these panels show where each catchment characteristics affects Q_5 and how that effect varies across the predictor range. Black lines indicate Brazilian biome boundaries. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

variability is limited in our sample (bare soil $\leq 3\%$; HDI $\leq 10\%$), thereby limiting breakpoint detectability; accordingly, we do not discuss them further.

3.3.1. Climate spatial variability and thresholds in streamflow response

Spatially (Fig. 4a–6a; S4a–S7a; S8b), the results show two regimes, separated by a consistent ϕ threshold across streamflow signatures (Table S7). Median thresholds range from 1.23 to 1.37, and the associated empirical 95% uncertainty intervals are narrow, generally spanning only a few hundredths, which indicates high threshold stability across the 10 random seeds. In water-limited domains (Caatinga and northern Cerrado), increases in ϕ pull Q downward: drier soils and high evaporative demand reduce rainfall–runoff conversion efficiency, depressing Q_{mean} and the lower quantiles, unless a sequence of storms elevates antecedent soil moisture (ASM), a primary modulator of floods (Sharma et al., 2018; Wasko et al., 2020; Livneh et al., 2024). These regions show system response to climate variability through unsaturated zone tracer profiles and groundwater fluctuations; recharge is typically low and

episodic (Scanlon et al., 2006). In this setting, river–aquifer connectivity (gaining vs. losing reaches) is a key lever for streamflow variability (Winter et al., 1998; Sophocleous, 2002; Schwaback et al., 2021). Thus, as ϕ rises and the water table declines, the likelihood of losing reaches and baseflow depletion increases. Conversely, recharge pulses and high ASM may raise the water table, strengthen baseflow, and reduce the dominance of losing reaches.

In contrast, along the humid corridor Amazon–to–South we observe reaches with $\psi_{\text{Aridity}} > 0$ (i.e., a slight increase in ϕ raises the quantiles) (Fig. 4a–6a; S4a–S7a; S8b). We propose that, in still energy-limited sectors of the Amazon (low ϕ), a small increase in ϕ coincides spatially with attenuating floodplain physiography, which sharpens the hydrologic response across the distribution. This is consistent with the strong river–floodplain attenuation documented for the Amazon Basin, where lateral storage and backwater delay and dampen flood waves when dominant (Paiva et al., 2013a, 2013; Paiva & Lima, 2024). In addition, moisture supply remains high thanks to the SALLJ, which channels vapor from the Amazon to southeastern South America and

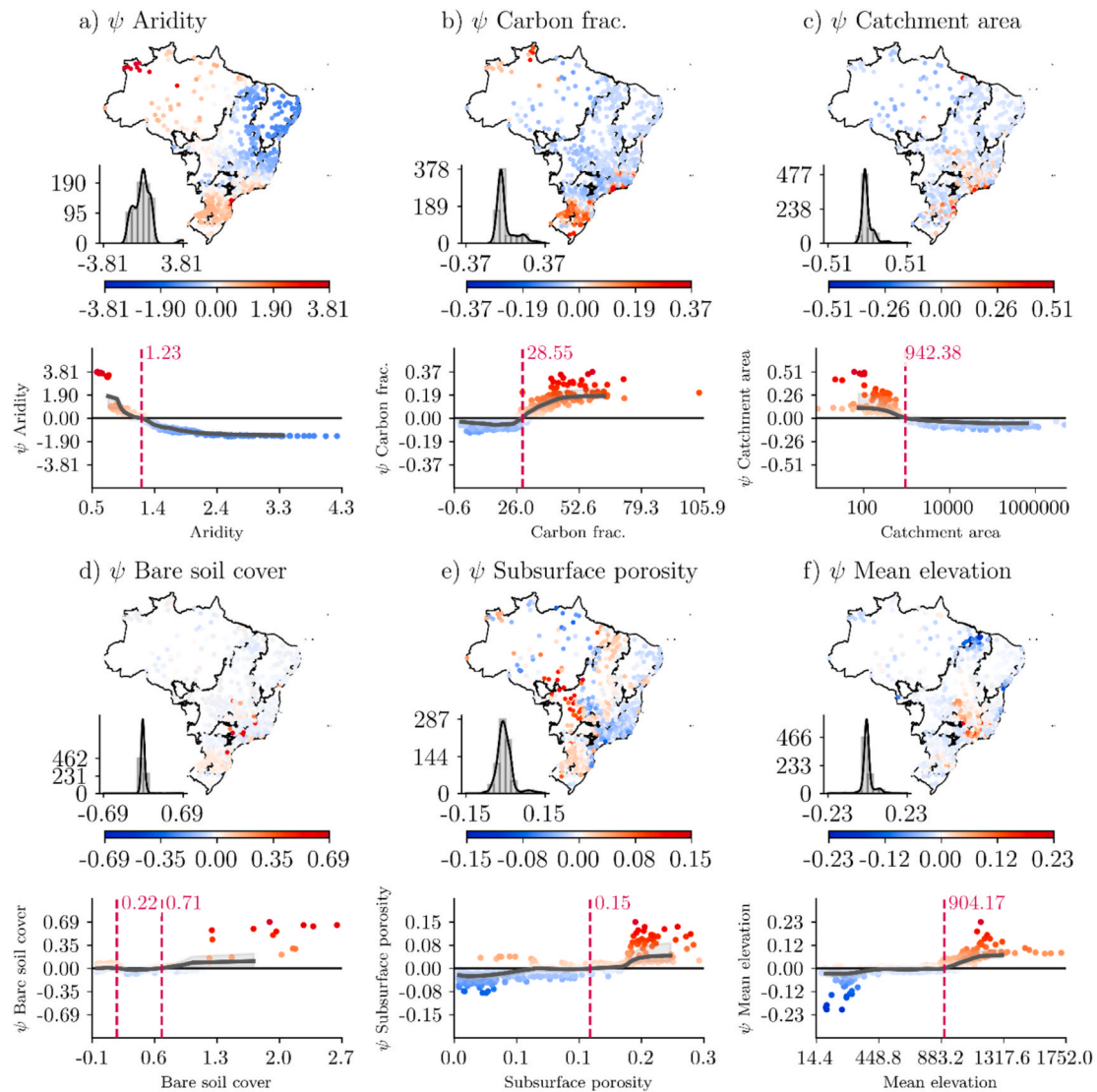


Fig. 5. Spatial SHAP patterns and potential thresholds for the six leading catchment characteristics of Q_{mean} across Brazilian catchments (remaining predictors in Figs. S6 and S11). For each catchment characteristics, the upper panel shows the spatial distribution of catchment-scale SHAP values (ψ_i), where warm colors indicate positive contributions to predicted Q_{mean} and cool colors indicate negative contributions; darker colors denote stronger effects. Insets summarize the ψ_i distributions, with $\psi_i = 0$ separating negative from positive effects. The lower panel shows the SHAP dependence plot, in which each point is one catchment, the black curve is the smoothed median response, and the gray band is the interquartile range. Horizontal zero lines mark the switch between negative and positive effects, and vertical dashed magenta lines mark potential breakpoints where the median SHAP response changes sign or slope. Together, these panels show where each catchment characteristics affects Q_{mean} and how that effect varies across the predictor range. Black lines indicate Brazilian biome boundaries. Abbreviation: “Carbon frac.” = soil carbon fraction. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

modulates organized rainfall and downstream extremes, sustaining precipitation sufficient for reduced attenuation/more efficient physiography to translate into larger Q (Montini et al., 2019). In the subsurface, the Amazon and other large Brazilian sedimentary basins host highly productive granular aquifers with large storage capacity (Satizábal-Alarcón et al., 2024), favoring baseflows; indeed, CAbra data indicate that most Brazilian rivers are baseflow-dependent, with BFI > 70% nationwide. In this context, positive ϕ —together with favorable soils and physiography and high precipitation—implies a shift in the atmosphere–surface balance that reduces lateral-storage/evaporation losses and increases the channelized fraction of streamflows. This becomes even clearer in the northernmost cluster, which is more exposed to that influence. Therefore, densely forested Amazon basins (cover $\geq 90\%$) show more sensitive streamflow responses to increases in ϕ . Such sensitivity is consistent with the fact that forest cover reductions decrease AET and tend to raise water yield and the quickflow fraction, given the change in vegetation use. By contrast, in the lower Paraná

Basin (25% forest, 35% grasslands, 25% croplands), although discharge volumes are similar, the same ϕ gradient is less pronounced. There, vegetation appears to exert a weaker influence on streamflow, whereas precipitation associated with the SALLJ (Smith et al., 2023), together with baseflow from the Guarani Aquifer (Sindico et al., 2018), may remain the dominant control.

SI it strongly shapes minimum streamflows because the timing between P and PET modulates recharge (Chagas et al., 2024), storage depletion, hydrological drought, and thus Q_1 – Q_5 . Spatially, Q_1 and Q_5 display similar patterns (Fig. 4b and S4b): $\psi_{\text{Prec. seasonality}} > 0$ concentrated in the country’s center (Aw/Cwa; summer-rain regimes), consistent with the SAMS and Brazil’s Köppen–Geiger climatology, where the SI reaches values near +1. Negative values appear north of the Amazon, in the northern Cerrado and along the northeastern coast (Af/Am), and in the South (Cfa/Cfb), in line with maps of the index itself (north, tendency toward -1 ; south, between -0.25 and $+0.25$). While increasing seasonality is associated with higher minima in the country’s

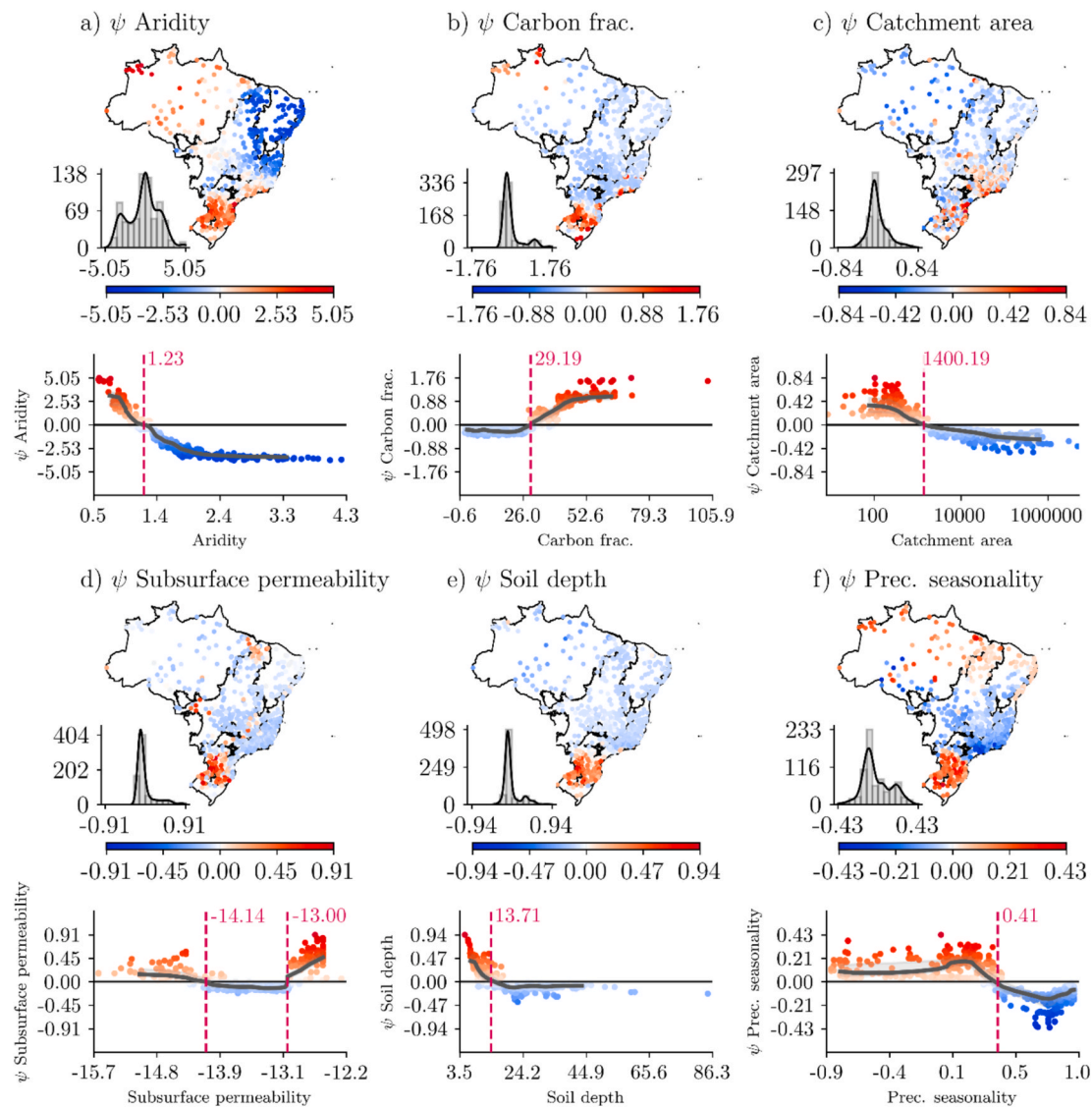


Fig. 6. Spatial SHAP patterns and potential thresholds for the six leading catchment characteristics of Q_{95} across Brazilian catchments (remaining predictors in Figs. S7 and S12). For each catchment characteristics, the upper panel shows the spatial distribution of catchment-scale SHAP values (ψ_i), where warm colors indicate positive contributions to predicted Q_{95} and cool colors indicate negative contributions; darker colors denote stronger effects. Insets summarize the ψ_i distributions, with $\psi_i = 0$ separating negative from positive effects. The lower panel shows the SHAP dependence plot, in which each point is one catchment, the black curve is the smoothed median response, and the gray band is the interquartile range. Horizontal zero lines mark the switch between negative and positive effects, and vertical dashed magenta lines mark potential breakpoints where the median SHAP response changes sign or slope. Together, these panels show where each catchment characteristics affects Q_{95} and how that effect varies across the predictor range. Black lines indicate Brazilian biome boundaries. Abbreviations: “Carbon frac.” = soil carbon fraction; “Prec. seasonality” = precipitation seasonality. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

center, observed hydrologic behavior there provides physical backing for the regional ψ signals: the nonlinear relationship between persistence of low streamflows in seasonal climates (Dralle et al., 2016) and very permeable and deep soils and granular aquifers (Wendland et al., 2015), with recharge on the order of 14–38% of annual P (Wendland et al., 2015; Oliveira et al., 2021). In this context, when the wet pulse is more concentrated, soil-moisture deficits are rapidly replenished (Westenbroek et al., 2010) and, in regions dominated by subsurface streamflows (Bartlett et al., 2016), thresholds for diffuse recharge are more easily crossed, yielding greater end-of-wet-season storage that sustains low streamflows.

In the northern Amazon and northern Cerrado, PET remains high year-round and the cool window for efficient recharge is weak; additionally, plutonic/volcanic terrains with fractured aquifers exhibit low productivity/storage (e.g., fractured–very low productivity, FU-VLP, or

fractured–low productivity/non-aquiferous, FU-LPoNA, see Satizábal-Alarcón et al., 2024). Along the northeastern coast, increasing seasonality deepens recession outside the peak: a short wet season and persistently high PET over a fractured crystalline basement with shallow bedrock limit effective storage (i.e., regional mean recharge is low, 8% of annual P; Costa et al., 2023), producing longer droughts and rapid recessions.

SI shows a coherent threshold structure across streamflow signatures (Table S7), with median thresholds around $SI = 0.4$ for Q_1 , Q_5 , Q_{95} , and Q_{99} (0.39, 0.37, 0.41, and 0.42, respectively), with narrow empirical 95% uncertainty intervals across the 10 random seeds, indicating a robust seasonal transition in streamflow response. For Q_1 – Q_5 , SI reduces streamflow below this range ($\rho = 0.27, 0.24$; $p < 0.05$; Fig. S14), but increases it above it (Fig. 4b and S8b). For Q_{95} – Q_{99} , the pattern reverses, $SI < 0.40$ is associated with higher peaks, while $SI \geq 0.40$ dampens them

(Fig. 7f, S13l, S14; Q_{95} , Q_{99} : $\rho = -0.17, -0.11$; $p < 0.05$). For Q_{mean} , the interval between -0.02 and 0.36 is associated with potential increases in mean streamflow, whereas $SI < -0.02$ is associated with decreases and $SI > 0.95$ with increases, suggesting a more nonlinear dependence on seasonality. This more complex behavior is expected because mean streamflow integrates both low-streamflow support and high-streamflow attenuation.

The sign contrast is physically plausible and consistent with prior literature: i. in more seasonal domains ($SI \geq 0.40$), a concentrated wet season elevates and sustains Q_1 – Q_5 (Van Loon, 2015; Dralle et al., 2016; Martínez-de la Torre & Miguez-Macho, 2019), whereas in areas with $SI < 0.40$ the unfavorable annual phase and smaller amplitude depress minima (Wunsch et al., 2024; Chagas et al., 2024); ii. in more seasonal domains ($SI \geq 0.40$), long dry spells depress antecedent moisture and, via intermittency, dampen Q_{max} (Wasko et al., 2020; Livneh et al., 2024), in contrast to areas with $SI < 0.40$ where frequently wet soils and events elevate Q_{max} . This threshold coincides with the Walsh–Lawler class break and maps onto Brazil’s geography (weakly seasonal north and south vs a more seasonal central belt), while the narrow uncertainty intervals further reinforce that the pattern reflects a coherent climatic control rather than spurious correlation.

3.3.2. Land use and land cover spatial variability and thresholds in streamflow response

Although many studies suggest that deforestation decreases ET and precipitation while increasing streamflow (D’Almeida et al., 2007; Zhang et al., 2017; Levy et al., 2018; Posada-Marín & Salazar, 2022), and that these effects may persist across scales (Zhang et al., 2017). However, the results presented here, interpreted through the theory of LSH, suggest that the relationship between forest fraction and the streamflow regime is nonlinear and multiscale. Three core results emerge: i. low streamflows are the most sensitive, with a breakpoint at 25% forest cover, above which Q_1/Q_5 decline (Figs. S9g, 4f) ii. Q_{mean} exhibits a threshold at 40% (Figs. S11h), consistent with the idea that up to this level, gains in infiltration/recharge/storage may coexist with higher ET, but beyond its mechanisms that reduce yield dominate in part of the contexts; and iii. for high streamflows (Q_{95} , Q_{99} ; Figs. S12o, S13o) forest cover matters less (with a threshold near 32%) because peaks are governed more by storm structure and synoptic, episodic hydraulic connectivity, and geomorphic controls that limit the share directly attributable to LULC. Unlike marginal correlations (Fig. S14) may appear positive simply because they capture a climatic co-gradient (wetter regions = more forest cover and more streamflow). Interpreting $\rho(X_{\text{forest}}, Y_{\text{FDC}})$ as causal is inappropriate under omitted-variable bias: $\text{Cov}(X_{\text{forest}}, \epsilon) \neq 0$ since climate jointly affects forest cover and FDC. Our ψ analysis reports interventional ψ_i contributions (Equation (4)), which are not causal per se but remove much of the climate, state and structure confounding and reveal regime-specific thresholds.

Our contribution is twofold: i. we establish interpretable thresholds (25–40%) with mechanistically plausible bases and well-constrained uncertainty ranges (Table S9), and ii. we do so at national scale while preserving hydroclimatic heterogeneity. These operational targets speak to the classic 20–40% threshold literature (Bosch & Hewlett, 1982; Costa et al., 2003; Levy et al., 2018; Hou et al., 2024; Hou & Wei, 2024), but extend it by revealing where and under which combinations of factors those thresholds manifest in Brazil. Despite a century of study, disagreement over forests’ influence on floods persists, partly because hydrology has at times prioritized process narratives over predictive validation and relied on chronological-pairing analyses that can understate flood effects relative to probabilistic, frequency-pairing framings (Nearing, 2020; Pham & Alila, 2024). Under this predictive-evidence perspective, our national, interventional analysis finds forest cover less influential for Q_{95} and Q_{99} than for Q_1 and Q_5 , indicating that upper-tail peaks are primarily set by storm physics, antecedent moisture, and geomorphic and hydraulic connectivity, with forest cover playing a secondary, context-dependent role.

Difference Vegetation Index in the summer $\text{NDVI}_{(\text{DJF})}$ shows possible thresholds around 0.70, above which Q_{mean} , Q_{95} and Q_{99} tend to increase (Figs. S11l, S12p, S13p). For Q_1 and Q_5 (Figs. S9n and S10p), $\text{NDVI}_{(\text{DJF})}$ shows two positive thresholds: lower values around 0.43–0.47 are associated with increases in low flows, and values above about 0.72 are also associated with higher minima. These patterns are consistent with moisture-stress transitions, whereby shifts across these NDVI ranges may reflect changes in vegetation and surface-moisture conditions that translate into higher streamflows. Low-streamflow signatures are particularly sensitive to vegetation changes (i.e., greater greenness can prolong the duration and frequency of low streamflows) helping explain why minima respond nonlinearly and, in some contexts, increase as $\text{NDVI}_{(\text{DJF})}$ drops but decline as canopy intensifies (Huang et al., 2022). Notwithstanding these patterns and the relatively narrow empirical uncertainty intervals (Table S9), $\text{NDVI}_{(\text{DJF})}$ had low relative importance in the model (Fig. S2) and non-significant correlations, and should therefore be interpreted as a secondary driver.

Grass cover shows asymmetric behavior across streamflow, with generally narrow empirical 95% uncertainty intervals around the estimated thresholds (Table S9). For Q_1 and Q_5 , a threshold near 26% is identified (Figs. S9f and S10g): below this level, minima tend to increase, whereas above it they potential to decrease. For Q_{mean} , Q_{95} and Q_{99} , thresholds near 37% are identified (Figs. S11p, S12l and S13i): below these levels, streamflows tend to decline, whereas above them they tend to increase. Mechanistically, expanding grass cover often raises evaporative and interception demand from shallow-rooted pastures and introduces compaction, so effective recharge does not keep pace, infiltration capacity falls, and a larger share of water is routed as quickflow—reducing minima while boosting peaks (and therefore the mean) via enhanced overland connectivity. In linear correlation, grass cover is negatively associated with streamflows overall (Fig. S14), but ψ reveals regime-specific sign reversals after marginalizing over other drivers.

3.3.3. Geology spatial variability and thresholds in streamflow response

Porosity follows the national spatial pattern (0–0.30) and emerges as influential for Q_1 , Q_5 and Q_{mean} streamflows and their thresholds (Fig. 4c, 5e and S9c). Spatially, Q_1 , Q_5 and Q_{mean} mirror this pattern: the ψ_{porosity} cluster is positive over the country’s central belt—coinciding with areas of higher porosity (0.30)—whereas values approach zero across Atlantic Forest basins and the far northern Amazon, which show the lowest porosities in our sample (< 0.10). Thresholds at 0.18 for Q_1 , Q_5 and Q_{mean} with narrow associated empirical uncertainty intervals (Table S8), indicate a sign change: below this value, porosity tends to depress streamflow ($-\psi$), while above them it tends to raise streamflow ($+\psi$). These breaks align with the spatial distribution, with porosity ≥ 0.18 characterizing regions where subsurface pathways are more developed and support baseflow, and porosity < 0.18 marking regions with limited storage and weaker low streamflows.

For upper quantiles, permeability supplants porosity in the rankings (Fig. 6d, S9f): for Q_{95} , $\rho = 0.11$ (permeability, $p < 0.05$) and $\rho = -0.18$ (porosity, $p < 0.05$); for Q_{99} , $\rho = 0.10$ (permeability, $p < 0.05$) and $\rho = -0.25$ (porosity, $p < 0.05$). Q_{95} and Q_{99} respond more to lateral transmissivity and subsurface connectivity during events than to bulk storage (Lehmann et al., 2007; Song et al., 2024). In more conductive systems, event recharge rapidly links hillslopes to channels via saturated subsurface stormflow, amplifying peaks. We also observe a higher concentration of catchments in southern Brazil (Fig. 6d, S6i and S8f) where permeability is associated higher streamflows, with saturated permeability $\log_{10}(k)$, where k is in m^2 , typically exceeding -13 . Across signatures, permeability shows generally narrow uncertainty intervals around the estimated thresholds across seeds (Table S8). For Q_1 and Q_5 , two candidate thresholds are apparent near $\log(k) = -13.50$ and -12.70 (Figs. S9p and S10o): below -13.50 m^2 , lower permeability is associated with reduced streamflows; between these thresholds, streamflows tend to increase; and above -12.70 , the relationship reverses again. For

Q_{mean} , a single threshold near $\log(k) = -13.0$ is identified (Fig. S11i), below which Q_{mean} tends to decrease and above which it tends to increase. For Q_{95} and Q_{99} (Figs. S6d and S13f) candidate thresholds occur near $\log(k) = -14.0$ and -13.0 , with streamflows tending to decrease between these values and increase outside this range.

HAND (Figs. S9–S13) a topographic proxy for local drainage potential, shows metric-specific candidate thresholds with relatively narrow empirical uncertainty intervals (Table S8). For Q_{mean} , a single threshold occurs near 103 m, below which Q_{mean} tends to decrease and above which it tends to increase. For Q_1 and Q_5 , the response is also nonlinear, with thresholds near 52–54 m and 163–171 m, and the correlations remain positive (Fig. S14). For Q_{95} and Q_{99} , two candidate thresholds emerge near 36–39 m and 93–95 m, suggesting a nonlinear response of high flows to local drainage potential. Spatially, $\text{HAND} < 115$ m spatially aligns with flatter, floodplain-rich and draining soils catchments where low relative elevation and shallow water tables imply low drainage potential and greater retention (Nobre et al., 2011). In contrast, higher HAND generally denotes steeper relief and a thicker vadose zone, favoring stronger drainage, weaker lateral storage connectivity, and faster runoff; soil moisture declines with relative elevation (Nobre et al., 2011). The higher upper thresholds observed for Q_1 and Q_5 suggests that baseflow support becomes limiting only in landscapes with still higher HAND, where disconnection from the water table may be more pronounced. Consistent with this interpretation, the coastal Atlantic Forest report deeper water tables (on the order of 80 m below ground) in high-HAND catchments (Almagro et al., 2021).

3.3.4. Soils spatial variability and thresholds in streamflow response

Soil carbon fraction shows a threshold near 30% across streamflow signatures, with generally narrow empirical uncertainty intervals (Figs. S9h, S10h, 5b, 6b, S13e; Table S8): below this level, streamflows tend to decrease, whereas above it they tend to increase. By contrast, correlations are uniformly positive across Q_1 – Q_{99} , capturing the overall monotonic tendency but not the breakpoint (Fig. S14). Spatially, it is difficult to isolate a clear relationship for the minima, but a cluster in southern Brazil stands out, where higher soil-carbon fractions align with higher streamflows (see Figs. S4–S8). The direction of effect is consistent with established soil physics: higher soil carbon fraction improves aggregation, porosity, water retention, and near-saturated hydraulic conductivity, thereby enhancing infiltration and potentially sustaining baseflow (Basche & DeLonge, 2019; Bargués-Tobella et al., 2024).

For low streamflows exhibit an upper threshold near 56% sand fraction, above which minima tend to increase, whereas lower sand fractions are generally associated with declines. These pattern is consistent with sand fraction relationships and higher recharge and baseflow in lighter-textured soils. For Q_{95} and Q_{99} , the pattern reverses around a threshold near 45% sand fraction: below this level, Q_{95} and Q_{99} tend to increase, whereas above it they tend to decrease, consistent with finer textures favoring surface runoff and peak amplification and sandier soils promoting infiltration and attenuation. For Q_{mean} , two thresholds emerge, although the one near 41% is more clearly defined. Beyond this value, the smoothed response remains close to zero, and catchments display both positive and negative effects across much of the observed range, while the overall monotonic tendency remains negative (Fig. S14). Spatial departures at Q_{95} and Q_{99} (Fig. S7a, S8a) are consistent with connectivity thresholds in shallow-to-bedrock (< 5 m) and volcanic settings: once storage fills and a perched water table forms, lateral transmissivity increases and subsurface stormflow activates, making sand fraction the dominant control on rare events in those clusters. These characteristics, coupled with high regional precipitation ($1,500 \text{ mm.yr}^{-1}$), plausibly account for local reversals of the general pattern. Overall, the estimated sand fraction thresholds are supported by generally narrow empirical uncertainty intervals, although some secondary breakpoints, especially for Q_{mean} , are less constrained (Table S8).

The bedrock-depth thresholds we detect map onto known hillslope hydrological mechanisms. For Q_1/Q_5 , two candidate thresholds emerge

near 9 and 25 m, with low-streamflow metrics tending to increase between these values and decrease outside this range. This pattern is consistent with a storage-use trade-off, in which intermediate critical-zone thickness supports subsurface storage and baseflow, whereas both very shallow and very deep profiles are less favorable to low-streamflow support. For Q_{mean} , the response is similarly non-monotonic, with a U-shaped pattern: Q_{mean} tends to increase below about 10 m, decrease between 10 and 26 m, and increase again above 26 m. For Q_{95} and Q_{99} , a shallower threshold near 14 m marks a reversal, with bedrock depths < 14 m associated with higher peaks and deeper profiles associated with attenuation. These patterns fit a non-linear storage–discharge framework in which shifts in available storage capacity and the activation of different streamflow pathways produce the observed sign-flips across the streamflow quantiles. Bivariate analyses confirm the overall inverse relationship between bedrock depth and streamflow quantiles (Fig. S14), highlighting the depth's strong control on higher streamflow generation. The estimated thresholds also appear relatively insensitive to random seeds, with generally narrow empirical uncertainty intervals (Table S8).

3.3.5. Topography spatial variability and thresholds in streamflow response

Topographic gradients modulate hillslope–channel connectivity and the fraction of rainfall water retained in the profile (Price, 2011). Accordingly, mean catchment slope (Fig. 4e and S4d) and mean elevation (Fig. 4d, S4e) emerge as influent controls on low streamflows (Q_1 : $\rho = 0.19$ and $\rho = 0.36$; Q_5 : $\rho = 0.23$ and $\rho = 0.37$; all $p < 0.05$), while for Q_{mean} (Fig. 5f) elevation remains the primary signal ($\rho = 0.18$, $p < 0.05$). Spatially, slope contributes positively along the Atlantic escarpments (Serra do Mar and Mantiqueira, 30%) and also on flat surfaces at the Caatinga–Cerrado transition (e.g., MATOPIBA, $< 1\%$), whereas increasing slope is associated with reductions in Q_1 , Q_5 and Q_{mean} in parts of the Amazon and southern Brazil. For mean elevation, positive signals cluster in montane areas (Mantiqueira/Serra do Mar, up to 2000 m), with negative signals most evident across the Amazon and north Cerrado.

Slope exhibits a U-shaped relationship with Q_1 , Q_5 and Q_{mean} , with candidate thresholds near 5% and 18% and generally narrow empirical 95% uncertainty intervals (Table S7; Figs. S9d, 4e, and S11o). Between these thresholds, increases in slope tend to reduce low and mean streamflows, whereas outside this range steeper slopes are associated with higher values. At the lower end, slopes $< 5\%$ are associated with higher low streamflows in catchments with larger drainable storage: Tertiary/Quaternary sedimentary deposits in depressions such as the northern Amazon, the Caiaçônia plateau, and the São Francisco mesas. Between 5 and 18%, slope tends to reduce low streamflows (shallower soils, limited lateral connectivity). Above 18%, low streamflows increase again ($\rho = 0.36$ – 0.37), as orographic effects become important in Atlantic Forest mountain belts (higher P, lower PET), together with metasedimentary covers and fracture or bedrock conductivity, help sustain dry-season baseflow ($\text{BFI} > 0.70$). For Q_{mean} , however, this pattern appears weaker than for the lower and upper streamflow signatures. For Q_{95} and Q_{99} , a single threshold occurs near 9%, above which high flows tend to increase and below which they tend to decrease (Figs. S12i and S13g). This pattern reflects the lateral redistribution imposed by relief, which switches hillslope–channel connectivity on and off; that transient connectivity controls runoff generation over time (Jencso et al., 2009; Mason et al., 2024), even though subsurface pathways are often predominantly hillslope-to-channel (Mason et al., 2024).

For mean elevation shows candidate thresholds, with generally narrow empirical 95% uncertainty intervals (Table S7). For Q_1 , Q_5 and Q_{mean} , thresholds occur near 485, 466, and 904 m, respectively, above which streamflows tend to increase and below which they tend to decrease (Figs. S9e, S4d and 5f). For Q_{95} , the response is U-shaped, with candidate thresholds near 375 and 1000 m; between these elevations, mean elevation tends to act as a depressor of high flows, whereas outside

this range it is associated with increases (Fig. S14k). Q_{99} broadly follows the same U-shaped pattern, with candidate thresholds near 576 and 1168 m, although the lower breakpoint appears more clearly defined than the upper one (Fig. S13m; Table S7). These patterns are consistent with lower-elevation settings (< 500 m) being warmer and more evaporative demanding ($PET > 5 \text{ mm} \cdot \text{d}^{-1}$), which favors storage drawdown. By contrast, intermediate elevations (500–900 m) show near-zero effects for Q_1 , Q_5 and Q_{mean} , whereas elevations ≥ 900 m (Araucaria ranges and the Serra do Mar escarpments/back-slopes on the eastern–southern plateau) is associated with higher streamflows due to orographic enhancement of P, $PET \leq 5 \text{ mm} \cdot \text{d}^{-1}$, and rainy-season groundwater rise that sustains baseflow in the dry season, aligning with results for montane environments in the Southeast (Chittolina et al., 2023).

Although larger drainage area might suggest higher streamflows, differences in catchment characteristics weaken a linear

area–streamflow relation (Saharia et al., 2017). Candidate breakpoints occur at 942 km^2 for Q_{mean} and $1300/1400 \text{ km}^2$ for Q_{95} and Q_{99} (Fig. 5c, 6c, and S13c), with generally limited empirical uncertainty, reaching at most about 71 km^2 (Table S7). Below these scales, mean and high streamflows tend to increase with area, whereas above them attenuation becomes systematic, consistent with the linear correlations for Q_1 – Q_{99} (Fig. S14). In other words, although correlations indicate an average decline of streamflow with area, the identified thresholds mark where the slope of the relationship changes, signaling a transition from a fast-response regime to one in which propagation and storage damp discharge per unit area (Marchi et al., 2010; Saharia et al., 2017; Paiva & Lima, 2024). These plateaus are consistent with literature values for the flash-to-fluvial transition, often reported around $1,000 \text{ km}^2$ (Marchi et al., 2010).

Results show two scale effects for low streamflows. For Q_1 (Fig. S9k)

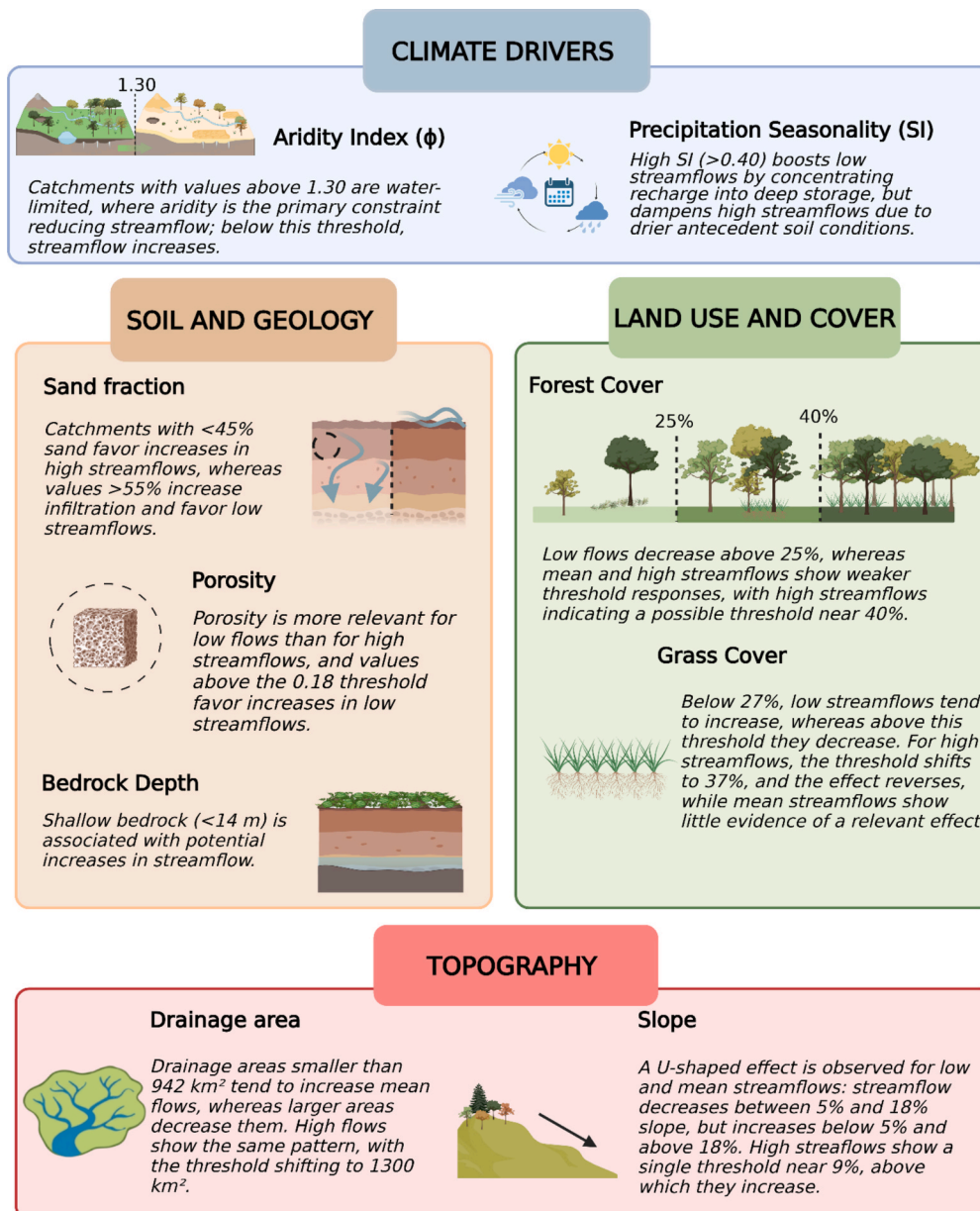


Fig. 7. Synthesis of the thresholds and relevant important catchments characteristic to hydrological transition zones identified across Brazilian catchments. The figure summarizes the most hydrologically interpretable candidate thresholds derived from SHAP dependence curves and spatial SHAP patterns, grouped into four domains: climate drivers, subsurface properties, land use and cover, and topography. The accompanying text in each panel summarizes the dominant streamflow condition under which the threshold was identified and its general interpretation for low, mean, or high streamflows. Created in BioRender. Brandão, A. (2026) <https://BioRender.com/m5wpjpr>.

exhibits positive ψ for both small and very large catchments, with candidate thresholds near 337 km² and 14,811 km², whereas Q_5 (Fig. S10i) shows positive ψ for areas < 576 km² and > 33,147 km². The headwater threshold for Q_5 (576 km²) likely reflects a compositional shift, whereby Atlantic Forest catchments underlain by fractured bedrock, with high HAND (mean 162 m), relatively high sand fractions (44%), and high baseflow indices (mean BFI = 0.87), display positive ψ for low flows, outweighing the negative ψ observed in drier counterparts (Figs. S6–S8). These thresholds are, however, less tightly constrained than those identified for several other predictors, with empirical 95% uncertainty intervals reaching nearly 5,100 km² for the upper breakpoints (Table S7).

3.3.6. Synthesis of thresholds and hydrological transition zones

Fig. 7 synthesizes the most stable, hydrologically interpretable, and most relevant inflection points identified across different streamflow signatures, grouped into climatic, subsurface, LULC, and topographic controls. Given their clearer physical consistency in the spatial SHAP patterns observed across Brazil, the selected thresholds should be interpreted as hydrological transition zones, that is, ranges of values beyond which the dominant streamflow response tends to change, rather than as fixed universal boundaries. As shown in Sections 3.3.1–3.3.5, both the position and the effect of these thresholds remain spatially contingent and signature-dependent.

The synthesis shows that the most recurrent and dominant threshold behavior is associated with aridity, SI, sand fraction, porosity, forest cover and grassland cover, drainage area, and slope. From a hydrological perspective, these variables represent the main dimensions through which Brazilian catchments organize streamflow: climatic water limitation, seasonal recharge concentration, infiltration and subsurface storage capacity, vegetation-mediated water partitioning, and the geomorphic efficiency with which water is transmitted from hillslopes to channels. Their threshold-shaped responses indicate that flow regimes do not vary smoothly across the full environmental gradient; instead, they often reorganize when important climatic, subsurface, or landscape conditions are exceeded. This helps explain why the same predictor may sustain low streamflows in one hydro-environmental context but attenuate them in another, or why controls on mean streamflow do not necessarily translate into controls on extremes.

This synthesis also helps fill an important gap in LSH in Brazilian studies. Previous continental and national studies have been highly effective in identifying which predictors are most important on average, but they have generally been less explicit about where predictor effects change sign, where nonlinear behavior becomes hydrologically significant, and which of these transitions are sufficiently stable to support interpretation beyond a purely statistical description. From a decision-making perspective, using an interventional method, the analysis evaluates the modeled contribution of each catchment characteristic after marginalizing over the background distribution of the remaining catchment characteristics, which helps reduce confounding from spatial co-gradients among climate, vegetation, soils, and topography. As a result, the identified thresholds are more informative for management than bivariate associations, while still requiring a non-causal interpretation. For example, aridity and SI thresholds help indicate where drought sensitivity and recharge timing may become dominant constraints on low streamflows; forest cover and grassland cover thresholds help identify ranges within which LULC change is more likely to alter low, mean, or high streamflows; and drainage area and slope thresholds help distinguish environments in which geomorphic organization favors attenuation rather than a faster hydrological response. These patterns do not prescribe universal management targets, but they provide reference zones that open the way for new studies that may support catchment planning, ecological flow protection, restoration prioritization, and climate adaptation strategies.

4. Conclusions

Understanding the complex, scale-dependent interactions between climate and catchment structure, as well as the critical thresholds that govern these relationships, remains a fundamental challenge in hydrology, mainly at scales where multiple processes operate simultaneously. The approach adopted here allowed us to move beyond average predictor importance and identify where the modeled effects of individual attributes changed sign, how these effects varied spatially, and which thresholds were sufficiently stable to support hydrological interpretation. RF performance was highest for Q_{mean} ($R^2 = 0.80$; KGE = 0.85), followed by Q_{95} ($R^2 = 0.71$; KGE = 0.77) and Q_5 ($R^2 = 0.63$; KGE = 0.70) whereas Q_1 ($R^2 = 0.63$; KGE = 0.63) and especially Q_{99} ($R^2 = 0.44$; KGE = 0.54) were less predictable, as expected for hydrologically more complex extremes. These metrics, together with the limited sensitivity to random-seed choice and the weak residual spatial autocorrelation, indicate that the inferred patterns are robust enough to support diagnostic analyses of attribute-streamflow relationships.

Our findings show that the influence of any single catchment characteristic is shaped by its co-occurrence with others. We note the critical thresholds at which a variable's role shifts from attenuating to amplifying streamflow responses, or vice versa. Among the 16 attributes grouped into state, structure, and climate categories, aridity is the dominant large-scale control on streamflows; with a transition near 1.23–1.37, beyond which drying exerts increasingly negative marginal effects on streamflows. However, this effect varies spatially: in regions like the Amazon and the South, increased aridity can enhance streamflow under similar rainfall regimes, while in drier areas it deepens streamflow reductions. Low streamflows were especially sensitive to precipitation seasonality, which showed a threshold near 0.40, as well as to porosity, slope, elevation, and vegetation cover. Another example is the forest cover that showed little relevance on high streamflows, but low streamflows were sensitive to LULC, with a 25% forest cover threshold above which low streamflows decline. For low streamflows, sand fraction also showed a threshold response, with values around 56% associated with increased infiltration and storage, thereby enhancing streamflow. For high streamflows, the threshold shifts to 45%, with a reversed relationship. This reversal is not spatially uniform; for example, in the South, sandy soils over shallow bedrock (<5 m) instead increase high streamflows. This marginal, threshold-based framework provides a diagnostic perspective that helps reconcile seemingly contradictory results in the hydrological literature, where analyses have often been limited to narrow ranges of catchment properties or streamflow regimes.

The results presented here describe marginal relationships rather than causal ones. While national datasets used to construct large samples and to represent extreme streamflows introduce uncertainty, the observed spatial coherence and hydrological consistency suggest that these are genuine physical signals rather than spurious correlations. Overall, Brazilian streamflow regimes appear to be structured by a set of physically interpretable thresholds that delineate transitions between regimes dominated by external forcing and those governed by catchment state and structure. This perspective helps reconcile mixed findings in the literature. In summary, this study provides both predictive and diagnostic value. The marginal thresholds identified here offer process-based reference points that can inform water resources planning and management. These findings have direct applications for improving drought resilience in agriculture, supporting targeted flood mitigation, and guiding the design of nature-based solutions. Further, this framework advances the understanding of large-sample hydrology by framing it as a set of interacting, threshold-governed regimes, contributing to ongoing efforts to enhance global water security.

CRedit authorship contribution statement

Abderraman R. Amorim Brandão: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology,

Investigation, Formal analysis, Data curation. **Admin Husic:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **André Almagro:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Dima-ghi Schwambach:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Paulo Tarso S. Oliveira:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Paulo Tarso Sanches de Oliveira reports financial support was provided by National Council for Scientific and Technological Development. Abderraman Brandao reports financial support was provided by Coordination for the improvement of Higher Education Personnel. Paulo Tarso Sanches de Oliveira reports financial support was provided by Fundação de Apoio ao Ensino, Ciência e Tecnologia do Estado de Mato Grosso do Sul. Paulo Tarso Sanches de Oliveira reports financial support was provided by Coordination for the improvement of Higher Education Personnel. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2026.135492>.

Data availability

The data used in this study is entirely based on hydrometeorological and catchment attributes from the CABra dataset, which are freely available online at <https://doi.org/10.5281/zenodo.4070146> (Almagro et al., 2021).

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